

Information System Audit

Program and Semester: BTECH CS CSE SEM VII		Pre-requisite Course :					
Academic Year: 2026-27		Credit Details:	L	T	P	C	H
			2	0	2	3	4
Faculty Name of Course: Dhanashree Kulkarni Contact Details: 9503055137/ dhanashree.huddedar@nmims.edu Office Hours: 8AM -4 PM		Faculty associated with the course: - 1.NA 2.NA					
Pre-Course Activity: • Discuss why you have opted for this elective subject. • Explore job descriptions of: ○ Information System Auditor ○ IT Risk Consultant ○ GRC Analyst ○ SOC Auditor ○ Compliance Analyst ○ Internal Auditor ○ CISA Professional ○ Compare salaries in India and globally. • Identify certifications: ○ CISA ○ ISO 27001 Lead Auditor ○ COBIT ○ CRISC ○ ISO 22301 ○ NIST CSF Using GenAI, explore the following: • Skills required for auditors. • Various types of audits under information system audit.							
Course link:							

1. Introduction to the Course

1.1 Importance of the Course

1.1.1 Domain Relevance:

The effectiveness of an information system's controls is evaluated through an information systems audit. An audit aims to establish whether information systems are safeguarding corporate assets, maintaining the integrity of stored and communicated data, supporting corporate objectives effectively, and operating efficiently. It is a part of a more general financial audit that verifies an organization's accounting records and financial statements. Information System Auditors today play a critical role in cybersecurity governance, cloud security assessments, compliance auditing, digital forensics, AI governance, privacy regulations (DPDP/GDPR), business continuity, and third-party risk management.

1.1.2 Industry Relevance:

Information systems are designed so that every financial transaction can be traced. In other words, an audit trail must exist that can establish where each transaction originated and how it was processed. Aside from financial audits, operational audits are used to evaluate the effectiveness and efficiency of information systems operations, and technological audits verify that information technologies are appropriately chosen, configured, and implemented.

Information System Auditors today play a critical role in cybersecurity governance, cloud security assessments, compliance auditing, digital forensics, AI governance, privacy regulations (DPDP/GDPR), business continuity, and third-party risk management.

1.2 Objectives of the Course:

Enhance the skills required to govern and control enterprise IT and perform an effective security audit.

2. Course Outcomes (CO), Mapping with Program Outcomes (PO), and Program Specific Outcomes (PSO)

2.1 Course Outcomes

- **CO1:** Identify and assess vulnerabilities and report on compliance and institutional controls.
- **CO2:** Gain a better understanding of IS audit and assurance guidelines, standards, and best practices.
- **CO3:** Learn how to govern and control an enterprise IT environment.
- **CO4:** 4. Understand the acquisition, development, testing and implementation of information systems
- **Program Outcomes(PO) the course contributed to:**
- **PO-1:** PO-1 An ability to apply knowledge of mathematics, science, and engineering for problem solving.
- **PO-2** An ability to research, design and conduct experiments, as well as to analyze and interpret data.
- **PO-3** An ability to design, implement, and evaluate a computer-based system, process, component, or program to meet desired needs.
- **PO-4** An ability to function effectively in teams to accomplish a common goal.
- **PO-5** An ability to identify, formulate and provide effective IT solution for engineering problems.
- **PO-6** An understanding of professional, legal, security and social issues and responsibilities
- **PO-7** An ability to communicate effectively with a range of audiences.
- **PO-8** The broad education is necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context.
- **PO-9** Recognition of the need for and an ability to engage in continuing professional development and self-learning.
- **PO-10** An ability to apply ethical principles in development of IT solutions.
- **PO-11** An ability to use the techniques, skills, and modern engineering tools necessary for developing effective IT solutions.
- **PO-12** An ability to identify and analyze user needs and take them into account in the selection, creation/integration, evaluation and administration of IT-based solutions.

2.2 Program Specific Outcomes (PSO):

- **PSO-1:** Demonstrate an ability to analyze and mitigate emerging cyber threats by applying cybersecurity principles and practices.
- **PSO-2:** Demonstrate an ability to recommend appropriate cybersecurity solutions considering business goals, industry standards and regulatory compliance requirements.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1		L			H									
CO2						H								M
CO3										H				
CO3														M

Mapping Levels: 1- High, 2-Medium, 3-Low

2.3 Student Outcomes (SO) (For ABET accredited Programs):

- **SO-1:**
- **SO-2:**

Teaching-learning methodology

2.4 Instruction Plan

Lecture No.	Topics	Teaching Method	Blooms Level Reference	Resources	COs mapped	Assessment and Evaluation
Unit 1 : Introduction						
1	Introduction to course	PPT, Whiteboard	Understand	TB1, RB2	CO1	Lab Submission and Class Test-1 and TEE
2	Management of the IS Audit Function				CO1	
3	ISACA IT Audit and Assurance Standards and Guideline				CO1	
4	Risk Analysis, Internal Control				CO1	
5	Performing an IS Audit Control Self-Assessment, The Evolving IS Audit Process				CO1	
Unit 2: Governance and Management of IT						
6	Corporate Governance, IT Governance (ITG),	Case Study	Understand	TB1, RB2	CO1	Lab Submission and Class Test-1 and TEE
7	Information Technology Monitoring and Assurance, Practices for Board and Senior Management		Analyse		CO2	
8	Information Systems Strategy, Maturity and Process Improvement Models,				CO2	
9	IT Investment and Allocation Practices, Policies and Procedures, Risk Management				CO2	
10	IS management Practices, IS Organizational Structure and Responsibilities				CO2	
11	Auditing IT Governance Structure and Implementation				CO2	

12	Business Continuity Planning, Auditing Business Continuity				C02	
13	Class Test 1					

Unit 3: IS Acquisition, Development, and Implementation

14	IS Acquisition, Development, and Implementation: Business Realization, Project Management Structure	PPT, Whiteboard	Evaluate	TB1, RB2	C02	Lab Submission and Class Test-2 and TEE
15	Project Management Practices, Business Application Development,				C03	
16	Business Application Systems, Alternative Forms of Software Project Organization,				C03	
17	Alternative Development Methods, Infrastructure Development/ Acquisition Practices,				C03	
18	Information Systems Maintenance Practices, System Development Tools and Productivity Aids				C03	
19	Process Improvement Practices, Application Controls, Auditing Application Controls, Auditing Systems Development, Acquisition and Maintenance.				C03	
20	Group discussion on IS Acquisition				C03	

Unit 4: IS Operations, Maintenance, and Support

21	Information Systems Operations , Information Systems Hardware	PPT, Whiteboard		TB1, RB2	C04	Lab Submission and Class Test-2 and TEE
22	IS Architecture and Software				C04	
23	IS Network Infrastructure, Auditing Infrastructure and Operation, Disaster Recovery Planning				C04	
24	Class Test 2					

25	Importance of Information Security Management				C04	
Unit 5: Protection of Information Assets						
26	Logical Access, Network Infrastructure Security, Auditing	PPT, Whiteboard	Understand/ Evaluate	TB1, RB2	C04	Lab Submission and Class Test-2 and TEE
27	Information Security Management Framework, Auditing Network Infrastructure Security, Physical Access				C04	
28	Environmental Exposures and Controls				C04	
29	Exposures and Controls, Mobile Computing.				C04	
Unit 6: Case Study						
30	Group discussion on real life case studies like PCI DSS, ISO 27001	PPT, Whiteboard	Understand/ Evaluate	TB1, RB2	CO4	Lab Submission and Class Test-2 and TEE

3. Assessment and Evaluation Scheme

	Internal Continuous Assessment (ICA) (50 Marks)				
	Class Tests (20 Marks)		Term-work (30 Marks)		
Assessment Component	Class Test-1	Class Test-2	Lab Performance (Report Submissions)	Audit Project (Group activity max 3)	Class Participation (Group Discussion, Class Assignments)
Marks	10	10	10	10	10

3.1 Internal Continuous Assessment (ICA) – 50 marks

Class test 1 and 2 (20Marks)

3.1.1 Two class tests will be conducted as per the academic calendar. It may be conducted offline for 10 marks each

3.1.2 Lab performance evaluation (10 marks)

Continuous assessment for laboratory experiments will be conducted. There are 10 practical, each carrying a weightage of 10 marks. At the end of the course, an average of total marks will be taken to obtain marks out of 20.

Discussion of your work with your peers is allowed. However, each student is expected to submit his/her original work. Submissions which are very similar will be marked zero. Assessment of the lab work will be carried out based on parameters like timely completion of lab work file, understanding of the experiment performed, originality in the work, involvement of the student, regularity, discipline etc. during the session. There is a 30% penalty on late submission.

3.1.3 Mini Project (10 marks) – Students will conduct an audit of an information system using internationally accepted audit methodologies (COBIT, ISO 27001, NIST CSF, CIS Controls, PCI DSS, DPDP Act where applicable).

The project should focus on identifying risks, evaluating existing controls, documenting audit findings, and proposing practical recommendations.

3.1.4 Class Participation (10 marks)- The faculty will ask some questions in every class based on the content being taught. The question could be asked to a chosen student or a student group (which is formed at the beginning of the semester). One mark can be given to the correct answer. The idea is to encourage students to pay attention in class and actively participate. These marks will be added in ICA class participation component. Class Assignments and group discussion will be held during the theory session.

3.2 Term End Examination (TEE) – (100 marks scaled down to 50)

A written examination of 100 marks for ISA duration will be conducted for the course as per the academic calendar.

3.3 Course Passing Criteria

3.3.1 ICA (50 marks) – No minimum marks

3.3.2 TEE (100 marks scaled to 50) – 40% required for passing

3.3.3 (ICA + TEE) (100 marks) - 40% required for passing

3.4 Assessments and Mapping to Course Outcomes

	Internal Continuous Assessment (ICA)					Term End Examination (TEE)
Course Outcomes	CT-1	CT-2	Lab Submissions	Audit Project (Group activity max 3)	Class Participation (Group Discussion, Class Assignments)	TEE
CO-1	Y		Y	Y	Y	Y
CO-2	Y		Y	Y	Y	Y
CO-3		Y	Y	Y	Y	Y

4. Laboratory details

The following 10 programming exercises will form the submission for laboratory coursework. Each programming exercise will contain 3 to 5 programs.

Lab No.#	List of Lab Exercises		Mapped CO
1	Create a CSA Questionnaire for College ERP	Week 2	CO1,CO3
2	Conduct Governance Audit of - Amazon - Google - Infosys using COBIT	Week 3	CO2
3	Design a complete BCP for NMIMS LMS outage.	Week 4	CO1,CO2

4	Create an Asset Register Risk Register, Threat Matrix using Excel.	Week 5	CO1
5	Audit a failed IT Project.	Week 6	CO3
6	Audit Acquisition/ SDLC documentation of a sample project.	Week 7	CO3,CO4
7	Perform Application Control Testing: Processing, Input, Output, Authorization, Logging, Exception Handling	Week 8	CO4, CO3
8	Conduct Disaster Recovery Simulation: student acts as CISO/Auditor /DBA/Network Admin Incident Manager and prepare DRP	Week 10	CO4, CO3
9	Perform ISO 27001 Gap Analysis.	Week 11	CO4, CO3
10	Select one framework PCI DSS,ISO 27001,NIST,CIS Controls, DPDP Act and perform a mock audit.	Week 12	CO4

5. Tutorial Plan

NA

6. Course Material

References and Lab Manuals would be uploaded on LMS by faculty every week.

7. GenAI Usage

7.1 Pre-class Activity:

- Generate ISO Controls
- Compare GDPR vs DPDP
- Map Controls to NIST

7.2 In-Class Activity:

Use GenAI for creating audit checklists for asset management.

Use GenAI for summarizing PCI DSS

Group discussion on real life incidents like:

- Equifax Data Breach
- SolarWinds Attack
- Colonial Pipeline
- CrowdStrike 2024 outage

- MOVEit breach
- AI-generated deepfake fraud
- Uber breach
- Capital One cloud incident

Students must validate all AI-generated content using authoritative standards before submission.

Academic Integrity Statement

Original work expected from students for all of the **assigned assessment work**. Copying in any form not acceptable and will invite strict disciplinary action. Evaluation of corresponding component will be affected proportionately in such cases. Plagiarism detection software will be used to check plagiarism wherever applicable. Academic integrity is expected from students in all components of course assessment.

*** - Only Teaching Method in the Instruction Plan for the course may vary for different faculty teaching the course, rest of the Course Policy Document will not change.**